Program : **Diploma in Polymer Technology**	
Course Code : **6092A**	Course Title: **Adhesives**
Semester : **6**	Credits: **4**
Course Category: **Open Elective**	
Periods per week: **4 (L:3 T:1 P:0)**	Periods per semester: **60**

Course Objectives:

- To impart basic terminologies related with adhesive bonding.
- To understand different surface preparation techniques adopted prior to adhesive bonding
- To understand different types of adhesives based on origin, function, nature of curing etc.
- To recognize various joint designs, testing methods related to adhesive bonding.

Course Prerequisites:

Topic	Course code	Course name	Semester
Basics of organic chemistry,Indroduction of polymers		Applied chemistry	I

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Summarize the significance of adhesive bonding and terminologies used in adhesive bonding	13	Understanding
CO2	Explain theories of adhesion as well as surface treatments adopted prior to adhesive bonding.	15	Understanding
CO3	Explain the adhesive classifications and adhesive composition	15	Understanding
CO4	Generalize various joint designs and testing methods employed in adhesive bonding	15	Understanding
	Series Test	2	

CO – PO Mapping:

Course Outcome	PO1	PO2	PO3	PO4	PO5	PO6	PO
CO1	3						
CO2	3						
CO3	3						
CO4	3						

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline:

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	**Summarize the significance and terminologies used in adhesive bonding**		
M1.01	Identify the concept and terminologies related to adhesive bonding.	3	Remembering
M1.02	Compile the advantages and disadvantages of adhesive bonding	3	Understanding
M1.03	Summarize the concept of surface tension, contact angle, wetting in adhesive bonding	4	Understanding
M1.04	Categorize the failure modes in adhesive bonding	3	Understanding
Contents: Adhesives - interphase - adhesive bonding - definition - functions of adhesives - advantages and disadvantages of adhesive bonding. Requirements of a good bond - proper choice of adhesive - good joint design - cleanliness - wetting - adhesive bonding process. Surface tension - Work of adhesion - contact angle - wetting - thermodynamic of adhesion - critical surface tension. Definition of failure modes - mechanism of bond failure.			
CO2	**Explain theories of adhesion as well as surface treatments adopted prior to adhesive bonding**		
M2.01	Explain the theories of adhesion	5	Understanding
M2.02	Recognize high energy and low energy surfaces	2	Remembering
M2.03	Identify surface treatments to be carried out for metals prior to adhesive bonding	4	Understanding
M2.04	Generalize surface treatments carried out in plastics before adhesive bonding	4	Understanding
	Series Test – I	1	

Contents: Theories of adhesion - mechanical theory - electrostatic theory - diffusion theory - weak boundary layer theory - Surface energy - Low energy surface - high energy surface. Surface treatments for metals - Degreasing - solvent cleaning - chemical treatment - priming. Surface treatments for plastics - surface cleaning - corona treatment - flame treatments - plasma treatments - Chemical etching - method for evaluating effectiveness of surface preparation- water - break test - contact angle test.			
CO3	**Explain the adhesive classifications and adhesive composition**		
M3.01	State different classifications of adhesives based on function, source, mode of application etc.	4	Understanding
M3.02	Summarize various adhesives used in adhesive industry	4	Understanding
M3.03	Differentiate structural and non - structural adhesives.	4	Understanding
M3.04	Identify composition in an adhesives used in industries	3	Understanding
Contents: Classification of adhesives - source - chemical composition - function - physical form - mode of application and setting - specific adherends or applications - examples. Structural and nonstructural adhesives. Types of adhesives - properties and applications - hot melt adhesive - pressure sensitive adhesive - cyanoacrylate adhesive - anaerobic adhesive - epoxy adhesives - phenolic adhesives - water based adhesives. Adhesive composition - binder - hardener - solvents - diluents - fillers - reinforcements - formulation - adhesives in electrical, automobile industry.			
CO4	**Generalize various joint designs and testing methods employed in adhesive bonding**		
M4.01	Indicate various stresses in a joint design	4	Understanding
M4.02	Explain the joint design criteria and typical joint designs used	4	Understanding
M4.03	Chart out the factors affecting performance of joint design	3	Understanding
M4.04	Summarize various testing methods used in adhesive bonding	4	Understanding
	Series Test – II	1	
Contents: Joint design - basic principles - types of stresses - tension - compression - shear - peel - cleavage - Joint design criteria - typical joint designs - stiffening joints - cylindrical joints - joints for plastics and elastomers - factors affecting performance of joint design. Testing of adhesives - Tensile - tear - cleavage - creep - fatigue - impact			

Text / Reference:

T/R	Book Title/Author
T1	P. Ghosh, Adhesive and Coating Technology, Tata McGraw-Hill, 2008 23.
T2	S.Ebnesajjad, Handbook of Adhesives and Surface Preparation: Technology, Applications and Manufacturing, William Andrew, 2010
R1	Skiests (Ed). Handbook of Adhesives, 3rd Ed., Van Nostrand Reinhold, 1990
R2	Shields, Handbook of Adhesives, Butterworths, 1984
R3	Pizzi (Ed), Wood Adhesives, Chemistry and Technology, Marcel Dekker 1983 5.
R4	Edward M. Petrie , Handbook of Adhesives and Sealants, Mcgraw-hill, 2000

Online Resources:

Sl. No.	Website Link
1	https://www.adhesives.org
2	https://www.sciencedirect.com
3	https://pubs.rsc.org